

Same-coordinate covariance under a separable Gibbs measure

Tide retrospective, laplace seabed

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1 Setting

The laplace seabed’s separable-potential module already proves that, under a product Gibbs measure $L(x, y) = U(x) + V(y)$, the covariance between a function of the first coordinate and a function of the second vanishes. This tide adds the diagonal companion: the covariance of two observables of the *same* coordinate is inherited unchanged from the corresponding one-dimensional marginal. The two facts together say the covariance matrix is block-diagonal for separable potentials — exactly the structural input the project’s Stage-3 multivariate law $\text{Cov}_t[\varphi, \psi] = (1/t)\langle \nabla \varphi, \Sigma \nabla \psi \rangle$ needs when Σ is itself separable. It is a deliberate step off the (now-complete) monomial cumulant ladder toward a different kind of result.

2 The thing formalised

For a separable potential and both marginal partition functions nonzero,

$$\text{Cov}_{2D}(f \circ \text{fst}, g \circ \text{fst}) = \text{Cov}_U(f, g), \quad \text{Cov}_{2D}(f \circ \text{snd}, g \circ \text{snd}) = \text{Cov}_V(f, g).$$

The Lean names are `gibbsCov_addSeparable_fst_fst_eq` and its second-coordinate twin, in a new leaf file `SeparableSameCoordCov.lean`.

3 Proof strategy

The covariance unfolds to $\langle fg \rangle - \langle f \rangle \langle g \rangle$. Each Gibbs expectation of a same-coordinate observable is computed by the existing separable-observable factorisation, pairing the observable with the constant 1 in the free coordinate: $\langle (fg)(z_1) \rangle_{2D} = \langle fg \rangle_U \cdot \langle 1 \rangle_V = \langle fg \rangle_U$, since the Gibbs expectation of the constant 1 is 1. Three such collapses (for fg, f, g) rewrite the 2D covariance into the 1D one; the identity then closes definitionally. This is the same collapse idiom the cross-covariance-vanishing proof already used, run three times on one coordinate.

4 Roadblocks and resolutions

The mathematics compiled on the first attempt. The one process lesson was about *where* to put the theorems. I first appended them to the core `AddSeparable.lean`; the proofs checked, but editing a widely-imported file forces a rebuild of every dependent, and those dependents surfaced their own pre-existing missing-header linter warnings — a cascade of noise unrelated to this tide. Relocating the additions to a new leaf file that imports `AddSeparable` leaves the core file untouched, so nothing downstream rebuilds and the tide’s build is warning-clean.

5 What was Mathlib, what was new

No new Mathlib input — the entire proof rests on the seabed’s own separable-observable factorisation and the constant-observable expectation. New here is the same-coordinate inheritance, which completes the block-diagonal covariance picture for separable potentials. No GPT consult was needed.